Applications of Duality: Revenue Management and Robust Optimization

How many tickets should an airline sell for each journey? How much should a factory produce when resource use is uncertain?

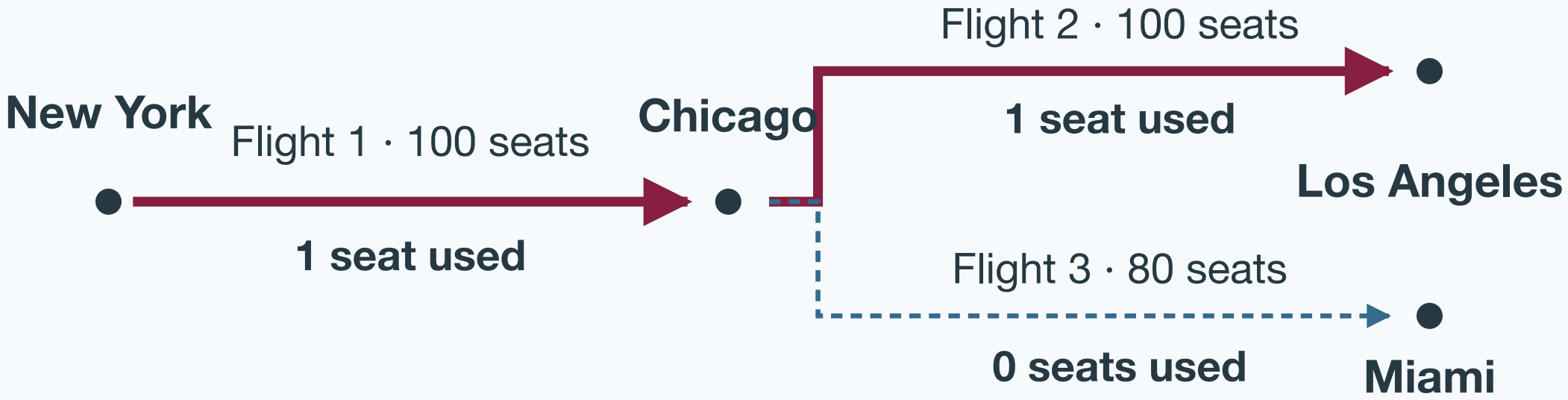
> **Two applications:** maximize airline ticket revenue; choose production quantities that satisfy the constraints for every data case we allow.

We will write each model, verify its numbers, and interpret its dual variables.

- Revenue management
- Robust optimization
- Uncertain capacities
- Proofs and further applications

One passenger may need seats on two flights

An airline has three direct flights with limited seats. Each direct flight is a **flight leg**; a passenger's complete journey is an **itinerary**.



Shown: one New York → Chicago → Los Angeles passenger uses one seat on each of flights 1 and 2.

> A connecting passenger uses one seat on **each** flight in the journey. Those seats could instead carry two direct-flight passengers.

The highest ticket fare need not give the most total revenue

Let x_j be the number of tickets sold for journey j. **Fare** means the ticket price per passenger for the complete journey.

j	Complete journey	Flights	Fare	Demand limit
1	New York → Chicago	1	$200	50
2	New York → Chicago → Los Angeles	1, 2	$400	100
3	New York → Chicago → Miami	1, 3	$350	70
4	Chicago → Los Angeles	2	$250	60
5	Chicago → Miami	3	$180	90

Demand limit: the maximum tickets we assume can be sold at that price.

> **One seat on each of flights 1 and 2:** one connecting ticket earns **$400**; two direct tickets earn **$200 + $250 = $450**, if both direct passengers want tickets.

Direct-ticket sales are demand limited. Choose the quantities of all five journeys together.

Choose ticket quantities subject to seat and demand limits

x_1, x_2, x_3: New York to Chicago, Los Angeles, Miami; the last two connect in Chicago. x_4, x_5: Chicago to Los Angeles, Miami.

$$
\begin{aligned}
\max \quad & 200x_1 + 400x_2 + 350x_3 + 250x_4 + 180x_5 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 \le 100 && \text{New York} \to \text{Chicago}, \\
& x_2 + x_4 \le 100 && \text{Chicago} \to \text{Los Angeles}, \\
& x_3 + x_5 \le 80 && \text{Chicago} \to \text{Miami}, \\
& 0 \le x \le (50, 100, 70, 60, 90).
\end{aligned}
$$

Each x_j is the number of tickets sold. Its upper bound is the demand limit for that journey. This LP treats these quantities as continuous.

> The objective is total ticket revenue in dollars. Each flight constraint counts the passengers occupying its seats.

A feasible allocation earns $57,100

Itinerary j	1	2	3	4	5
Accepted quantity	50	40	10	60	70
Demand limit	50	100	70	60	90

$$\begin{aligned}\text{Leg 1: } & 50 + 40 + 10 = 100,\\ \text{Leg 2: } & 40 + 60 = 100, \quad \text{Leg 3: } 10 + 70 = 80,\\ \text{Revenue: } & 200(50) + 400(40) + 350(10) + 250(60) + 180(70) = 57{,}100.\end{aligned}$$

This proves feasibility and computes the revenue. Optimality requires an upper bound of the same value.

Assign one dual variable to each upper bound

Use $p_1, p_2, p_3 \geq 0$ for the three leg capacities, and $u_1, \ldots, u_5 \geq 0$ for the five demand limits.

$$
\begin{aligned}
\min \quad & 100p_1 + 100p_2 + 80p_3 \\
& \quad +50u_1 + 100u_2 + 70u_3 + 60u_4 + 90u_5 \\
\text{s.t.} \quad & p_1 + u_1 \geq 200, \quad p_1 + p_2 + u_2 \geq 400, \\
& p_1 + p_3 + u_3 \geq 350, \quad p_2 + u_4 \geq 250, \quad p_3 + u_5 \geq 180, \\
& p, u \geq 0.
\end{aligned}
$$

Each fare is bounded by the total value assigned to its legs and its demand limit.

Complementary slackness determines these prices

At $x = (50, 40, 10, 60, 70)$, demand limits 2, 3, and 5 have positive slack. Therefore $u_2 = u_3 = u_5 = 0$.

All five accepted quantities are positive, so their dual constraints are active:

$$p_3 = 180, \quad p_1 + p_3 = 350, \quad p_1 + p_2 = 400, \quad p_1 + u_1 = 200, \quad p_2 + u_4 = 250.$$

$$p = (170, 230, 180), \qquad u = (30, 0, 0, 20, 0).$$

An active primal constraint does not by itself force a positive dual variable.

Check feasibility and the matching upper bound

Itinerary	Dual left side	Fare
1	170 + 30 = 200	200
2	170 + 230 = 400	400
3	170 + 180 = 350	350
4	230 + 20 = 250	250
5	180	180

$$100(170) + 100(230) + 80(180) + 50(30) + 60(20) = 57{,}100.$$

All dual variables are nonnegative, all five constraints hold, and the dual value equals the feasible revenue. Both solutions are optimal.

Compare a new fare with the legs it consumes

The **bid price** of a leg is its capacity dual variable. Here $p = (170, 230, 180)$.

New request	Resource value	Example decision
Leg 1 only	$170	$160 is below the threshold.
Leg 2 only	$230	$240 exceeds the threshold.
Legs 1 and 3	$170 + $180 = $350	$350 gives equality.

> This is a marginal planning rule. It assumes the request can replace the current allocation and that the current prices remain applicable.

The dual variables on demand limits measure the benefit of having more requests of those existing types.

Check the $170 opportunity cost by changing bookings

Current allocation: $x = (50, 40, 10, 60, 70)$. Consider **one additional** leg-1-only request paying 180.

Change	Capacity consequence	Revenue change
Reduce itinerary 3 by 1	Frees one seat on legs 1 and 3	–350
Increase itinerary 5 by 1	Uses the freed leg-3 seat	+180
Accept the new request	Uses the freed leg-1 seat	+180

$$\text{Lost value of the old allocation} = 350 - 180 = 170, \qquad \text{net gain} = 180 - 170 = 10.$$

A fare can exceed one local fare and still be unattractive

With $p = (170, 230, 180)$, a new connection on legs 1 and 2 pays 390. Its resource value is $170 + 230 = 400$.

> The fare is below the combined bid price, even though it exceeds the fare of either local product.

At equality, the local LP calculation is indifferent. Neither equality nor a positive reduced profit guarantees unlimited capacity or unlimited future demand.

> As capacity and demand change, solve the planning LP again and update its prices.

A fixed plan must survive every allowed scenario

A nominal LP uses one estimated data set. A robust LP chooses a plan that satisfies its constraints for every data set in a specified uncertainty set.

> **Order of decisions:** choose the production quantities first; the uncertain coefficients are realized afterwards.

The uncertainty set is part of the model. Robust feasibility protects against its scenarios, not against events outside it.

> Start with a small set of scenarios, then use duality when there are infinitely many.

Two products with uncertain resource consumption

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & (2+z_1)x_1 + (1+z_2)x_2 \le 8, \\ & x_1 + x_2 \le 5, \qquad x_1, x_2 \ge 0. \end{aligned}$$

The uncertain increases satisfy $z_1, z_2 \ge 0$, $z_1 + z_2 \le 1$. The same x must work for **every** such z.

> At the nominal scenario $z = (0, 0)$, the optimal plan is $x = (3, 2)$, with profit 13.

The nominal plan fails two allowed scenarios

For the plan $x = (3, 2)$, resource consumption is $(2 + z_1)3 + (1 + z_2)2$.

Scenario	Consumption	Capacity 8
$(0, 0)$	8	Feasible
$(1, 0)$	11	Infeasible
$(0, 1)$	10	Infeasible

Protecting against only the estimated coefficients gives no guarantee for the other allowed scenarios.

Here three scenarios describe the entire uncertainty set

Every allowed z is a convex combination of $(0,0), (1,0), (0,1)$. Consumption is linear in z, so checking those vertices is sufficient.

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \le 8, \quad 2x_1 + 2x_2 \le 8, \\ & x_1 + x_2 \le 5, \quad x \ge 0. \end{aligned}$$

$$x^* = (2,2), \qquad \text{profit} = 10.$$

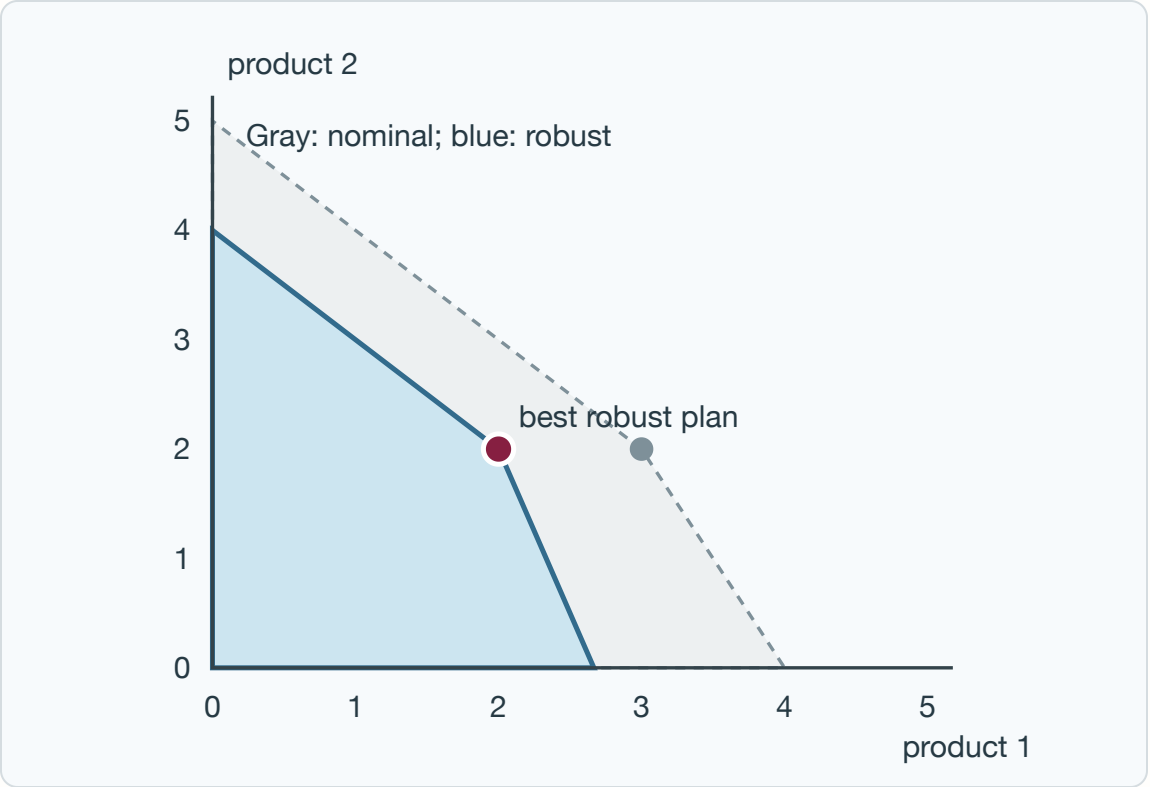
The original nominal row is implied by these rows and nonnegativity.

More coefficient uncertainty can change the best plan

Use $0 \le z_1, z_2 \le 1$, $z_1 + z_2 \le \Gamma$, where $\Gamma \in [0, 2]$ is the uncertainty budget.

> The model remains $\max 3x_1 + 2x_2$, with capacity 8, $x_1 + x_2 \le 5$, and $x \ge 0$.

At $\Gamma = 0$, no deviation is allowed. At $\Gamma = 2$, both coefficients may reach their largest values.



Budget 1: plan (2, 2), profit 10.

Fix the plan, then ask for its worst consumption

For the budget $\Gamma = 1$, the uncertain extra consumption is

$$\begin{aligned} \max_{z_1, z_2} \quad & x_1 z_1 + x_2 z_2 \\ \text{s.t.} \quad & z_1 + z_2 \le 1, \qquad z_1, z_2 \ge 0. \end{aligned}$$

Here x is fixed data for the inner LP. The adversary chooses z, not the production quantities.

> Put the whole budget on the larger of x_1, x_2: worst extra consumption is $\max\{x_1, x_2\}$.

The dual replaces infinitely many tests by three inequalities

The inner LP has one constraint. Give it a dual variable $q \geq 0$:

$$\min_{q} \; q \quad \text{s.t. } q \geq x_1, \quad q \geq x_2, \quad q \geq 0.$$

> Strong duality makes this minimum equal to the worst extra consumption.

$$2x_1 + x_2 + q \leq 8, \qquad q \geq x_1, \quad q \geq x_2.$$

These finite LP constraints are equivalent to protecting against every allowed z. The auxiliary variable q bounds the worst extra consumption.

The same construction works for a polyhedral coefficient set

Let the uncertain coefficient vector $a \in \mathbb{R}^n$ lie in the nonempty set $U = \{a : Da \leq d\}$. The robust row is $a^\top x \leq \beta$ for every $a \in U$.

$$\max_{a: Da \leq d} a^\top x \quad \longleftrightarrow \quad \min_{\lambda \geq 0} \{d^\top \lambda : D^\top \lambda = x\}.$$

$$D^\top \lambda = x, \qquad d^\top \lambda \leq \beta, \qquad \lambda \geq 0.$$

The coefficients a are unrestricted in sign, so the dual coefficient condition is an **equality**. A robustly feasible row has a finite inner maximum.

Why the new dual variables certify every scenario

Suppose $D^\top \lambda = x$, $d^\top \lambda \le \beta$, and $\lambda \ge 0$. For any a with $Da \le d$,

$$a^\top x = a^\top D^\top \lambda = (Da)^\top \lambda \le d^\top \lambda \le \beta.$$

One feasible dual certificate bounds every allowed scenario at once. Strong duality supplies the reverse implication.

Full equivalence proof in the appendix.

Uncertain capacities require a different worst-case direction

$$\max\ 40x_1 + 30x_2 \quad \text{s.t. } 2x_1 + x_2 \le b_1, \quad x_1 + 2x_2 \le b_2, \quad x \ge 0.$$

The capacities satisfy $80 \le b_1 \le 100$, $60 \le b_2 \le 80$, and $b_1 + b_2 \ge 150$.

> A fixed production plan must work for every allowed capacity pair. Smaller capacities make these $\le$ constraints harder to satisfy.

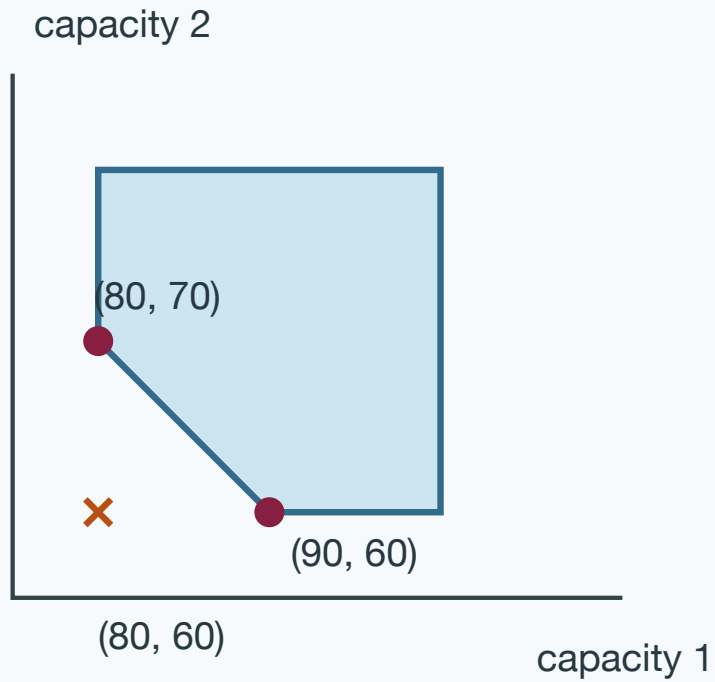
Therefore row i requires $a_i^\top x \le \min_{b \in U} b_i$.

Minimize each right-hand side separately

For row 1, the minimum is $b_1 = 80$, attained at $(80, 70)$. For row 2, the minimum is $b_2 = 60$, attained at $(90, 60)$.

$$2x_1 + x_2 \leq 80, \qquad x_1 + 2x_2 \leq 60.$$

A plan must satisfy row 1 in its worst scenario and row 2 in its worst scenario. Those scenarios need not be identical.



The two row minima need not occur in the same scenario.

Solve the nominal and robust production problems

Model	Capacities	Optimal quantities	Profit
Nominal	$(100, 80)$	$(40, 20)$	2200
Robust	$(80, 60)$	$(100/3, 40/3)$	$5200/3 \approx 1733.33$

At the robust optimum, solve $2x_1 + x_2 = 80$ and $x_1 + 2x_2 = 60$. The nonnegative dual variables are $(50/3, 20/3)$.

$$80(50/3) + 60(20/3) = 5200/3.$$

The feasible production plan attains its dual upper bound. Protection reduces the best profit by about 466.67.

Correlation does not help this fixed-plan RHS model

The pair $(80, 60)$ is excluded by $b_1 + b_2 \geq 150$. Nevertheless, the separate minima remain 80 and 60.

> Removing that pair does not change the robust feasible set: each row must still withstand its own minimum.

Correlation matters when it changes a row's worst value, or when uncertain coefficients jointly enter the same constraint. It does not automatically improve every robust model.

Allowing decisions to change after observing the scenario would be a different model.

A shared RHS budget saturates at one full deviation

Let $b_i = \bar{b}_i - \hat{b}_i z_i$, where $0 \le z_i \le 1$, $\sum_i z_i \le \Gamma$, and $\Gamma \ge 0$.

For one row i, the adversary can concentrate its budget there: $\max z_i = \min\{1, \Gamma\}$.

$$a_i^\top x \le \bar{b}_i - \hat{b}_i \min\{1, \Gamma\}.$$

For $\Gamma \ge 1$, every row already has to survive its full individual deviation. Increasing the budget further does not change this fixed-plan RHS model.

Calculate the budget effect rather than assuming it

Use $\bar{b} = (100, 80)$, $\hat{b} = (20, 20)$ in the production LP $\max 40x_1 + 30x_2$.

Γ	Robust capacities	Optimal quantities	Profit
0	(100, 80)	(40, 20)	2200
0.5	(90, 70)	$(110/3, 50/3)$	$5900/3$
1	(80, 60)	$(100/3, 40/3)$	$5200/3$
1.5 or 2	(80, 60)	$(100/3, 40/3)$	$5200/3$

Budgeted coefficient uncertainty can continue changing after Γ=1. Pure RHS uncertainty in this fixed-plan form cannot.

The uncertain quantity determines the inner LP

Uncertainty	Worst-case test	Duality gives
Coefficients in a ≤ row	$\max_{a \in U} a^\top x \le \beta$	An upper bound on consumption.
RHS in a ≤ row	$a^\top x \le \min_{b \in U} b_i$	A lower bound on available capacity.
Returns in a maximized objective	$\max_x \min_{r \in U} r^\top x$	A guaranteed return for the chosen portfolio.

Full RHS dual construction, including its signs.

Portfolio and routing examples.

Interpret a dual solution; then use duality as a modeling tool

Revenue management: compute resource bid prices, add them across the resources used, and verify displacement numerically.

Robust optimization: fix a decision, optimize over the uncertainty set, write the inner dual, and substitute its finite constraints.

Keep units, sign conventions, decision timing, and the uncertainty set explicit.

Next, large-scale LP uses dual variables to find improving columns and violated constraints. Cutting-stock column generation belongs to that next resource.

Appendix: complete derivations and additional applications

These pages complete the derivations and retain further continuous-LP applications.

- Revenue dual and marginal rules
- Robust equivalence proofs
- RHS uncertainty signs
- Portfolio and corridor models
- Dispatch and capacity value

Derive the general booking dual

Let $A_{ij} = 1$ when itinerary j uses leg i, and 0 otherwise. Let fares be f, capacities C, and demand limits d.

$$\begin{aligned} \max \quad & f^\top x \\ \text{s.t.} \quad & Ax \le C, \\ & 0 \le x \le d. \end{aligned} \qquad \begin{aligned} \min \quad & C^\top p + d^\top u \\ \text{s.t.} \quad & A^\top p + u \ge f, \\ & p, u \ge 0. \end{aligned}$$

For any feasible pair, $f^\top x \le p^\top A x + u^\top x \le p^\top C + u^\top d$.

> The max primal has ≤ rows, so the min dual has nonnegative variables. Nonnegative primal quantities give ≥ dual constraints.

Every complementary pair in the booking example

Primal $x = (50, 40, 10, 60, 70)$; dual $p = (170, 230, 180)$, $u = (30, 0, 0, 20, 0)$.

$$C - Ax = (0, 0, 0), \qquad d - x = (0, 60, 60, 0, 20).$$

$$A^\top p + u - f = (0, 0, 0, 0, 0).$$

$$p_i(C_i - A_i x) = 0, \qquad u_j(d_j - x_j) = 0, \qquad x_j((A^\top p)_j + u_j - f_j) = 0.$$

Positive primal slack forces the paired dual variable to zero. Zero primal slack merely permits a nonzero dual variable.

Recover the allocation directly from the certificates

The positive dual variables p_1, p_2, p_3, u_1, u_4 require the corresponding primal rows to be active:

$$x_1 = 50, \quad x_4 = 60, \quad x_2 + x_4 = 100, \quad x_1 + x_2 + x_3 = 100, \quad x_3 + x_5 = 80.$$

$$x_2 = 40, \qquad x_3 = 10, \qquad x_5 = 70.$$

> Check every demand bound and nonnegativity afterwards. Complementary equations alone do not replace feasibility checks.

The same booking basis gives a precise validity region

Keep the same demand limits and the active bounds $x_1 = 50, x_4 = 60$. For capacities $C = (C_1, C_2, C_3)$, solve the three active leg equations:

$$x_2 = C_2 - 60, \qquad x_3 = C_1 - C_2 + 10, \qquad x_5 = C_3 - C_1 + C_2 - 10.$$

$$\begin{aligned} 0 \le C_2 - 60 \le 100, \\ 0 \le C_1 - C_2 + 10 \le 70, \\ 0 \le C_3 - C_1 + C_2 - 10 \le 90. \end{aligned}$$

Within this region, revenue equals $170C_1 + 230C_2 + 180C_3 + 2700$. Outside it, reoptimize.

A bid price is the reduced-cost test for a new request

Add a new nonnegative variable t, fare r, and resource-use column a. The old solution with $t = 0$ remains feasible.

$$\bar{r} = r - p^{\top} a.$$

> For this maximization problem, $\bar{r} \leq 0$ preserves the old dual bound. If $\bar{r} > 0$, the old certificate fails for the new column and primal simplex can test an improving move.

A positive reduced profit is a direction test; degeneracy or capacity restrictions can limit its feasible step.

Robust coefficient rows: prove both directions

Let $U = \{a : Da \le d\} \neq \varnothing$. Fix x and a finite bound β.

If there is $\lambda \ge 0$ with $D^\top \lambda = x$ and $d^\top \lambda \le \beta$, then for every $a \in U$, $a^\top x = (Da)^\top \lambda \le d^\top \lambda \le \beta$.

Conversely, if $a^\top x \le \beta$ for every $a \in U$, the feasible inner LP has a finite optimum. LP strong duality gives an optimal $\lambda \ge 0$ with $D^\top \lambda = x$ and $d^\top \lambda \le \beta$.

The nonempty uncertainty-set assumption avoids treating an impossible scenario model as a useful guarantee.

Budgeted coefficient uncertainty with individual caps

Let $a = \bar{a} + \hat{a} \odot z$, with $\hat{a}_j \geq 0$, $0 \leq z_j \leq 1$, $\sum_j z_j \leq \Gamma$, and $x \geq 0$. Here $\odot$ means componentwise multiplication.

$$\max_z \sum_j \hat{a}_j x_j z_j \quad \text{s.t.} \ \sum_j z_j \leq \Gamma, \quad 0 \leq z_j \leq 1.$$

Use $\lambda \geq 0$ for the budget row and $\mu_j \geq 0$ for each upper bound.

$$\min \ \Gamma\lambda + \sum_j \mu_j \quad \text{s.t.} \ \lambda + \mu_j \geq \hat{a}_j x_j, \quad \lambda, \mu \geq 0.$$

The budgeted robust row is a finite LP constraint

For $(\bar{a} + \hat{a} \odot z)^\top x \le \beta$ in every budgeted scenario, introduce the inner dual variables:

$$\bar{a}^\top x + \Gamma\lambda + \sum_j \mu_j \le \beta, \qquad \lambda + \mu_j \ge \hat{a}_j x_j, \qquad \lambda, \mu \ge 0.$$

> Here the deviations for several coefficients compete inside one constraint. Increasing Γ beyond 1 can allow more coefficients to deviate together.

For the two-product example, $\bar{a} = (2, 1)$, $\hat{a} = (1, 1)$, and $\beta = 8$. At $\Gamma = 2$, the robust row is $3x_1 + 2x_2 \le 8$.

Construct the RHS dual with the correct signs

Let $U = \{b : Db \le d,\ b \ge \ell\} \neq \varnothing$. For row i, solve $\min_{b \in U} e_i^\top b$, where e_i has a 1 at position i.

Write both constraint groups as ≥: $-Db \ge -d$, $b \ge \ell$. Treat b as unrestricted; its lower bound is an explicit row.

$$\begin{aligned} \max \quad & -d^\top \lambda + \ell^\top \mu \\ \text{s.t.} \quad & -D^\top \lambda + \mu = e_i, \qquad \lambda, \mu \ge 0. \end{aligned}$$

The minus signs on D and d come from reversing the ≤ rows before forming a minimization dual.

A lower bound on every possible capacity

The robust row $a_i^\top x \le b_i$ for every $b \in U$ is equivalent to the existence of $\lambda_i, \mu_i \ge 0$ satisfying

$$-D^\top \lambda_i + \mu_i = e_i, \qquad a_i^\top x \le -d^\top \lambda_i + \ell^\top \mu_i.$$

Indeed, for every $b \in U$,

$$b_i = -\lambda_i^\top D b + \mu_i^\top b \ge -d^\top \lambda_i + \ell^\top \mu_i \ge a_i^\top x.$$

Conversely, strong duality supplies such variables if the nonempty inner LP has a finite minimum and the robust row holds.

Why different worst scenarios still constrain one fixed plan

$$\left[\forall b \in U :\ a_1^\top x \le b_1 \text{ and } a_2^\top x \le b_2\right]$$

$$\iff \quad \left[\forall b \in U :\ a_1^\top x \le b_1\right] \text{ and } \left[\forall b \in U :\ a_2^\top x \le b_2\right].$$

Each row must hold for all scenarios. The scenario producing row 1's minimum may differ from the scenario producing row 2's minimum.

> Coupling the right-hand sides matters only if it changes at least one of these separate minima, when all decisions are fixed before uncertainty is observed.

Verify the robust production optimum without a solver

The robust LP is $\max 40x_1 + 30x_2$ subject to $2x_1 + x_2 \le 80$, $x_1 + 2x_2 \le 60$, $x \ge 0$.

$$\frac{50}{3}(2x_1 + x_2) + \frac{20}{3}(x_1 + 2x_2) = 40x_1 + 30x_2 \le \frac{5200}{3}.$$

The plan $(100/3, 40/3)$ makes both inequalities active and attains the bound.

> For $t = \min\{1, \Gamma\}$, capacities are $(100 - 20t, 80 - 20t)$, quantities are $(40 - 20t/3, 20 - 20t/3)$, and profit is $2200 - 1400t/3$.

Portfolio protection under box uncertainty

Let $x_j \geq 0$ be the fraction invested in asset j, with $\sum_j x_j = 1$. Returns satisfy $\bar{r}_j - \hat{r}_j \leq r_j \leq \bar{r}_j + \hat{r}_j$.

$$\max_{x \geq 0,\, \mathbf{1}^\top x = 1} \; \min_{r \in U} r^\top x.$$

Because each $x_j \geq 0$, the smallest return chooses every $r_j = \bar{r}_j - \hat{r}_j$. Thus the model becomes

$$\max \; (\bar{r} - \hat{r})^\top x \quad \text{s.t. } \mathbf{1}^\top x = 1, \quad x \geq 0.$$

This simple box model need not diversify. Its constraints and uncertainty set must capture the intended investment restrictions.

A numerical portfolio comparison

Asset	Nominal return	Maximum drop	Worst return
A	8%	6%	2%
B	6%	1%	5%

With nonnegative fractions summing to 1 and no other restrictions, the nominal model puts everything in A; the robust box model puts everything in B.

> The guaranteed return increases from 2% to 5%, while the nominal return decreases from 8% to 6%. Neither model promises what will happen outside its specified scenarios.

Which uncertainty sets preserve a linear program?

Set	Geometry	Robust counterpart
Box	Coordinate bounds	LP
Budget / absolute-value bounds	Polyhedron after linearization	LP
Ellipsoid / Euclidean ball	Curved boundary in general	Usually a norm constraint, not an LP

$$\max_{\|a-\bar{a}\|_2 \le \rho} a^\top x = \bar{a}^\top x + \rho \|x\|_2.$$

An ellipsoid is not generally polyhedral. The Euclidean-norm formula is a contrast showing where the LP dualization argument changes.

Correlated delays: a continuous routing model

Two corridors each have two arcs. Corridor A takes 5+5 minutes nominally; B takes 6+6. A shared incident budget allows one unit of disruption.

$$z_A, z_B \ge 0, \quad z_A + z_B \le 1.$$

Each A arc gains $4z_A$ minutes; each B arc gains $3z_B$. Let $t \in [0,1]$ be the fraction of flow sent on A, and $1-t$ the fraction on B.

$$\min_{0\le t\le 1}\Big[10t + 12(1-t) + \max_{z_A+z_B\le 1}\big(8tz_A + 6(1-t)z_B\big)\Big].$$

Dualizing the adversary permits a split flow

The worst surcharge is $\max\{8t, 6(1-t)\}$. Introduce its upper bound q:

$$\begin{aligned} \min \quad & 12 - 2t + q \\ \text{s.t.} \quad & q \geq 8t, \quad q \geq 6(1-t), \quad 0 \leq t \leq 1. \end{aligned}$$

$$8t = 6(1-t) \Rightarrow t = 3/7, \qquad \text{value} = 102/7 \approx 14.57.$$

Sending everything on either single corridor gives worst-case time 18. The continuous model improves the guarantee by splitting flow. It does not select one path.

The general robust flow reformulation

Let x be a continuous feasible flow, $\bar{c}$ the nominal arc costs, and Δ the matrix mapping incidents $z \geq 0$ to added costs. Let $Bz \leq h$ describe incident budgets.

$$\min_{x \in P} \left[\bar{c}^{\top} x + \max_{z \geq 0,\, Bz \leq h} (\Delta^{\top} x)^{\top} z \right].$$

$$\min_{x \in P,\, \pi \geq 0} \quad \bar{c}^{\top} x + h^{\top} \pi \quad \text{s.t. } B^{\top} \pi \geq \Delta^{\top} x.$$

P contains flow conservation and continuous capacity bounds. The extra robust constraints need not preserve integral solutions. Binary single-path restrictions would create an integer-programming model.

Another price system: two-node generation dispatch

Node 2 needs 100 units of energy. Generator 1 costs 20 per unit, generator 2 costs 50; at most 60 units can move from node 1 to node 2.

$$\begin{aligned} \min \quad & 20g_1 + 50g_2 \\ \text{s.t.} \quad & g_1 - f = 0, \quad g_2 + f = 100, \quad 0 \le f \le 60, \quad g_1, g_2 \ge 0. \end{aligned}$$

Set $g_1 = f$, $g_2 = 100 - f$. Every extra transferred unit saves 30, so $(g_1, g_2, f) = (60, 40, 60)$, with cost 3200.

This transport-style model omits the voltage-angle equations of a physical power-flow model.

Derive and verify the dispatch dual

Use unrestricted p_1, p_2 for the two balance equalities and $-\eta \le 0$ for the upper bound $f \le 60$.

$$\begin{aligned} \max \quad & 100p_2 - 60\eta \\ \text{s.t.} \quad & p_1 \le 20, \quad p_2 \le 50, \quad -p_1 + p_2 - \eta \le 0, \quad \eta \ge 0. \end{aligned}$$

$$p_1 = 20, \quad p_2 = 50, \quad \eta = 30, \qquad 100(50) - 60(30) = 3200.$$

The optimal dual variables give the marginal demand costs at the nodes and the value of relaxing the transfer limit.

Capacity has value until the next change of basis

Replace 60 by $U \geq 0$. For $0 \leq U \leq 100$, use $f = U$, $g_1 = U$, and $g_2 = 100 - U$.

$$C(U) = 5000 - 30U \quad (0 \leq U \leq 100), \qquad C(U) = 2000 \quad (U \geq 100).$$

> One extra unit of transfer capacity saves 30 while the line is the limiting resource. Beyond 100, it adds no value in this model.

Expanding from 60 to 100 saves 1200 per modeled operating period. A cost of 25 per unit of added capacity gives a modeled net benefit $1200 - 25(40) = 200$.

A smaller booking model with an active demand limit

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + 6x_3 \\ \text{s.t.} \quad & x_1 + x_3 \le 4, \quad x_2 + x_3 \le 4, \\ & 0 \le x_1, x_2 \le 4, \quad 0 \le x_3 \le 3. \end{aligned}$$

Products 1 and 2 are local bookings; product 3 uses both legs. A connection earns 6 versus 3+2 from the two locals it replaces.

$$x = (1, 1, 3), \qquad p = (3, 2), \quad u = (0, 0, 1), \qquad \text{primal value} = \text{dual value} = 23.$$

Separate capacity prices from demand prices

For the preceding two-leg model, keep connection demand equal to 3. With capacities C_1 and C_2, the same allocation pattern is

$$x_1 = C_1 - 3, \quad x_2 = C_2 - 3, \quad x_3 = 3, \qquad 3 \le C_1, C_2 \le 7.$$

$$V(C_1, C_2) = 3C_1 + 2C_2 + 3.$$

Instead keep capacities at 4 and let connecting demand be D. For $0 \le D \le 4$,

$$x = (4 - D, 4 - D, D), \qquad V(D) = 20 + D.$$

> The capacity prices are 3 and 2; the connection-demand price is 1. They answer different perturbation questions.

Competition for time slots also produces resource prices

Jobs A and B each need one slot; one peak and one off-peak slot are available.

Value	Peak	Off-peak
A	10	6
B	9	8

Let $x_{it} \geq 0$ allocate job i to slot t. Maximize the displayed values times x_{it}, subject to one unit per job and at most one unit per slot.

> A in peak and B in off-peak earns 18; the reversed assignment earns 15. Slot-capacity dual variables measure the opportunity cost of that competition.

The assignment lecture will develop the full primal–dual algorithm.

A local price is one input to an investment decision

Scenario	Hours	Value per hour	Annual value
High load	1000	30	30000
Medium load	2000	10	20000
Low load	5760	0	0

> One unit of capacity has gross annual value 50,000 under these scenarios. Multiplying a peak-hour price by all hours would overstate its value.

Keep units and validity ranges explicit. Reoptimize after substantial changes, and account for operational restrictions missing from the LP.

Different optimal dual solutions can give different prices

Degeneracy or redundant constraints can produce multiple optimal dual vectors. Each remains a valid certificate, but a single reported coordinate may not describe both directions of a perturbation.

> Check the proposed direction by changing the data and solving again. The directional change in value is more reliable than interpreting an arbitrary reported coordinate in isolation.

At a corner of the optimal-value function, one-sided marginal values can differ.

Study guide and next connection

Be able to: construct the booking dual, verify the allocation and certificate, explain displacement, and turn a worst-case linear problem into a finite robust LP.

Keep distinct: coefficient and RHS uncertainty; capacity and demand prices; a fixed plan and a plan that adapts; continuous split flow and a single path.

Read the complete derivations here before applying a formula to a new sign convention. Large-scale LP next uses reduced costs to select columns and separation to select constraints.